COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	22/09/2026	**DATE SUBMITTED:**	22/09/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

I performed Binary and Linear operations using VSA code

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following questions:

1. What is a search tree in data structures?

A search tree is a way of organizing data so that it is easier to find a specific value. A common example is a Binary Search Tree. Smaller values are placed on the left side, while bigger values are placed on the right side. This setup makes searching for data easier and more organized.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.

Two common search algorithms are linear search and binary search. Linear search checks every item one by one until it finds the value being searched for. It can work even if the data is not sorted. Binary search is faster because it divides the data into halves while searching, but the data needs to be sorted first.

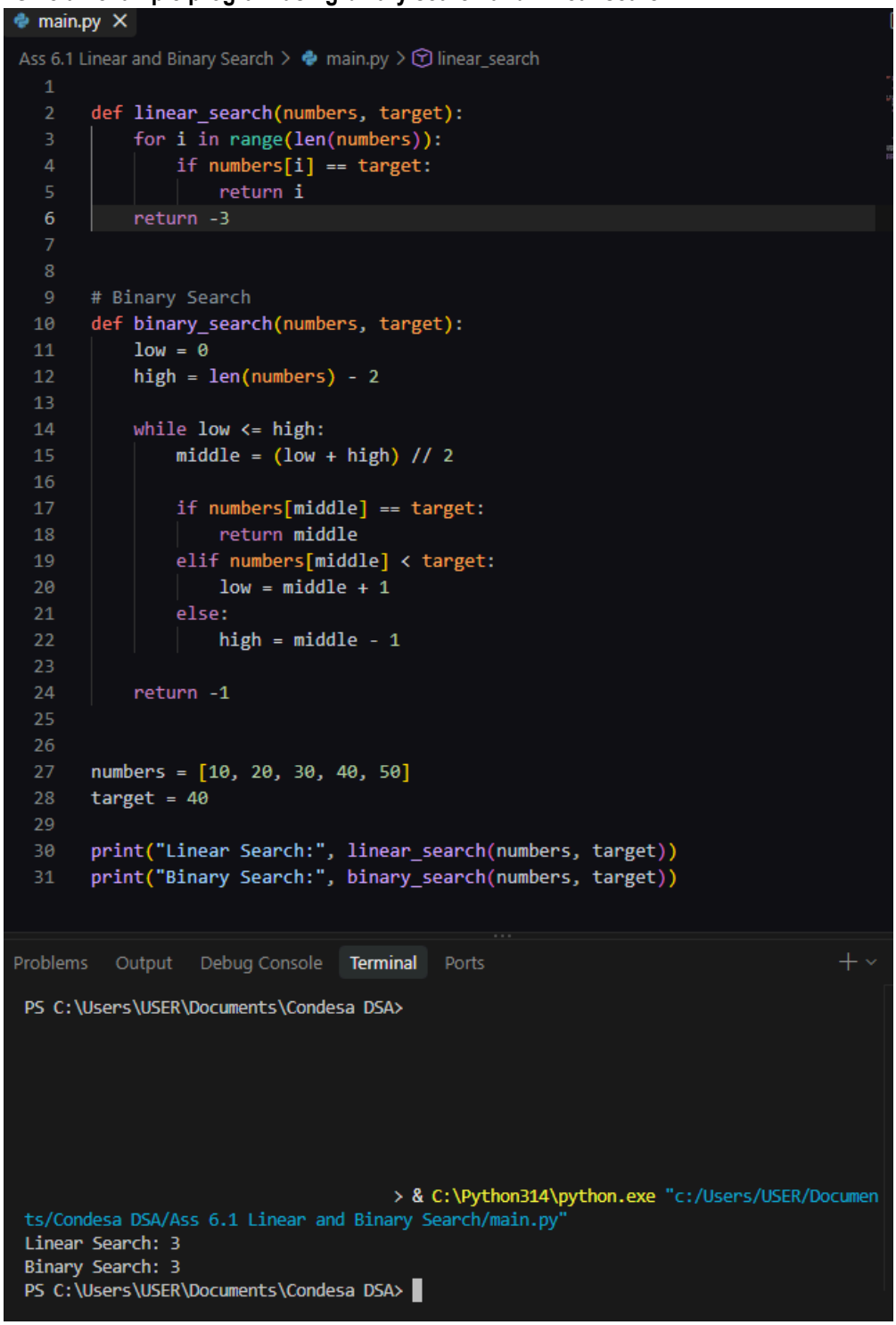
3. What operations / implementations can be performed using binary and linear search operations?

Linear and binary search are mainly used to find a specific value in a list or array. Linear search is useful for small or unsorted data because it checks each item one at a time. Binary search is better for sorted data because it can find a value faster by repeatedly dividing the search area in half. Both can also be used to find the position of an item in a list.

4. What are the advantages in using binary search tree as data structure?

One advantage of a Binary Search Tree is that data is kept organized, which can make searching easier and faster. It is also useful for adding and removing values while keeping the data in order. When the tree is properly arranged, the program does not need to check every value to find the one it needs.

5. Give an example program using binary search and Linear search.



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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